1. What is Spring Data JPA?
It is a Spring module that simplifies database access using JPA by reducing boilerplate code via repositories.

2. Difference between JPA and Hibernate
JPA is a specification, Hibernate is its implementation.

3. What is an Entity in JPA?
A POJO class mapped to a database table using @Entity annotation.

4. Hibernate Session vs EntityManager
Session is Hibernate-specific, EntityManager is JPA standard.

5. Persistence Context
It is a cache that manages entity lifecycle and tracks changes.

6. save() vs saveAndFlush()
save() persists later, saveAndFlush() immediately writes to DB.

7. ID generation strategies
AUTO, IDENTITY, SEQUENCE, TABLE.

8. @Transactional
Manages transaction boundaries and ensures atomic operations.

9. Dirty Checking
Automatically updates DB when entity state changes.

10. First-level cache
Default cache in persistence context per session.

11. LAZY vs EAGER
LAZY loads on demand, EAGER loads immediately.

12. N+1 Problem
Multiple queries issue; solved using fetch joins or EntityGraph.

13. Cascade types
Defines propagation like PERSIST, MERGE, REMOVE.

14. mappedBy vs @JoinColumn
mappedBy defines inverse side, @JoinColumn defines owner.

15. orphanRemoval
Deletes child when removed from parent.

16. JPQL vs Native Query
JPQL uses entity model, native uses SQL.

17. Criteria API
Type-safe dynamic query building.

18. Pagination & Sorting
Handled via Pageable interface.

19. Projections
Fetch partial data using DTO or interface.

20. Complex joins
Use JPQL or native queries.

21. Optimistic vs Pessimistic locking
Optimistic uses versioning, pessimistic locks DB rows.

22. @Version
Used for optimistic locking.

23. Transaction handling
Managed via proxies and AOP.

24. Flushing
Syncs persistence context with DB.

25. get() vs load()
get() hits DB immediately, load() uses proxy.

26. Improve performance
Use caching, batching, indexing.

27. Debug slow queries
Enable SQL logs and analyze execution plans.

28. Large data handling
Use batch processing and pagination.

29. Real issues
N+1, lazy loading exceptions, performance.

30. Relationship design
Prefer unidirectional, avoid deep nesting.